Haofei Xu

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EDUCATION

Washington University in St. Louis August 2025 – Present

Ph.D. in Computational and Data Science

University of Michigan, Ann Arbor August 2022 – May 2024

Master of Science in ***Information****, concentrating in* ***User-Centered Agile Development*** **GPA:** 4.0/4.0

University of Illinois at Urbana-Champaign August 2018 – May 2022

Bachelor of Science in ***Social Psychology****, Minors in* ***Computer Science*** *and* ***Applied Statistics*** **GPA:** 3.93/4.0

RESEARCH EXPERIENCE

Sentiment Analysis and User Profile Classification on 2024 US Election Tweets Ann Arbor, MI

Hackathon Project December 2024

- **Developed and fine-tuned** sentiment analysis models for the 2024 US Election Twitter dataset, leveraging **Logistic Regression, LSTM, and DistilBERT** to classify tweets, achieving a **98.54% accuracy** with DistilBERT.
- Implemented **large-scale data processing pipelines** for 1M+ tweets, applying **VADER**-based sentiment labeling, **text preprocessing (tokenization, lemmatization, stopword removal)**, and **feature engineering** (**BoW**, **embeddings**) to enhance model performance.
- Designed and conducted user profile analysis, **classifying** 13.5K+ Twitter profile images using **zero-shot learning** with a transformer model (google/siglip-so400m).
- Performed statistical analysis **(Chi-Square, ANOVA, Spearman Correlation, Regression)** to identify **significant** relationships between profile image types and sentiment, contributing insights into online behavior and political discourse.

A Large-Scale Dataset to Examine Political Discourse in Local Governance Ann Arbor, MI

Research Assistant August 2023 – February 2025

- Designed and deployed a scalable **ETL data pipeline** on **AWS** using **Terraform** to automatically extract agendas, minutes, and videos from over 500 Michigan municipalities daily. Utilized **Selenium**, **Playwright**, and other Python libraries for custom scraping scripts from government websites, YouTube, and Vimeo.
- Executed meticulous **data cleaning** processes, standardizing city names and other variables, which improved **data accuracy** by 95% and ensured consistency across datasets, crucial for subsequent analysis and visualization tasks.
- Applied advanced **text data analysis** and created over 50 impactful **data visualizations**, translating intricate datasets into clear and actionable insights, facilitating a deeper understanding of local political impacts, and driving informed discussions within the team.
- Implemented **BERTopic** for **topic modeling** analysis on over 5k public comments from city council meetings from 20 Michigan cities, totaling approximately 100k words. Conducted inference to extract and categorize key topics, effectively enhancing understanding of community sentiments. Tracked and improved model performance, achieving a 90% accuracy rate in topic classification.
- Presented research at the **Midwest Speech and Language Days** to researchers and professors from universities in the Midwest. Highlighted the use of a **scalable machine learning pipeline**, including **fine-tuned RoBERTa** models and novel **PublicSpeak algorithms**, to identify and classify public remarks from city council meetings. **Demonstrated key findings** on participation dynamics, seasonal variations in public engagement, and frequently discussed topics, sparking interdisciplinary dialogue and potential collaborations.

Accessibility in Higher Education: A Comparative Analysis between the US and China Ann Arbor, MI

Independent Research November 2023 – April 2024

- Conducted **computational analysis** of university websites for compliance with WCAG 2.0 AA standards using **automated testing tools** and **statistical methods** to identify significant differences in accessibility violations between the U.S. and China.
- Performed **on-site field studies** at universities in both countries, observing facilities and **conducting surveys and interviews** with students and staff to assess accessibility challenges.

- **Presented research findings** to an audience of librarians, professors, and peer researchers. **Delivered actionable recommendations**, including infrastructure improvements, awareness campaigns, and web accessibility training, aimed at enhancing accessibility standards and practices in higher education.
- Secured funding and completed the project under the **UM Library Mini Grant** program, demonstrating strong grant-writing and project management skills.

PUBLICATIONS

1. Ge, C., Zhang, J., Xu, H., Krupnikov, Y., Bednar, J., & Tomkins, S. (2025). What does the public want their local government to hear? A data-driven case study of public comments across the state of Michigan. *arXiv preprint arXiv:2507.18431*.

PRESENTATIONS

1. **Conference on Language Modeling,** Montreal, QC, Canada — October 2025
   - NLP 4 Democracy Workshop
2. **Midwest Speech and Language Days 2024,** Ann Arbor, MI — April 2024
   - A large-scale dataset to examine political discourse in local governance

WORK EXPERIENCE

J&K Scientific LLC — San Jose, CA

Data Engineer — June 2024 – Present

- Designed and implemented tools using **Python** or **Node.js** with **AWS Lambda** to automate the extraction of publicly available data, employing techniques like **Selenium**, and **BeautifulSoup4** to navigate and collect information efficiently. Reduced data delivery time by 98%, responding to over 500 requests daily and enhancing business intelligence to support informed decision-making.
- Orchestrated an auto-scalable **data pipeline** using **Docker**, **AWS S3, Step Functions**, **Lambda**, and **Batch** to automate regular web data collection tasks. Managed scheduling and execution of tasks to ensure timely and accurate retrieval of data, handling over 10 million data points efficiently and reliably. Reduced operational costs by 70% and data retrieval time by 50%.
- Engineered a **data warehousing and mining** solution utilizing **DynamoDB** and **Hadoop** architecture. Used DynamoDB as the central database for middleware to serve real-time data to the front end.
- Implemented change data capture and synchronization to consolidate data into an S3-based data lake with **Apache Iceberg**. Leveraged **Trino** as the query engine and incorporated Apache **Spark** for advanced analytics, supporting operational insights and decision-making.